

■ NIFTY-50 AI Investment Intelligence Platform

Comprehensive Technical Evaluation & Portfolio Optimisation Report

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1. Introduction

This technical document presents the framework, architecture, and empirical performance metrics of our NIFTY-50 AI Investment Intelligence Platform. The system is designed to transform raw historical stock price series into actionable asset allocation signals, combining statistical learning, Modern Portfolio Theory (MPT), and risk management. The model targets a 5-day forward forecasting horizon to match active trading and rebalancing cycles, leveraging both predictive modeling and robust risk assessment.

2. Exploratory Data Analysis & Feature Engineering

Quantitative strategies rely heavily on stable feature representations. We analyze the underlying distribution of daily returns across individual stocks in the index. Non-normality (leptokurtosis) and volatility clustering are dominant features, particularly evident during crisis periods (e.g., the 2008 Subprime collapse and the 2020 COVID-19 crash).

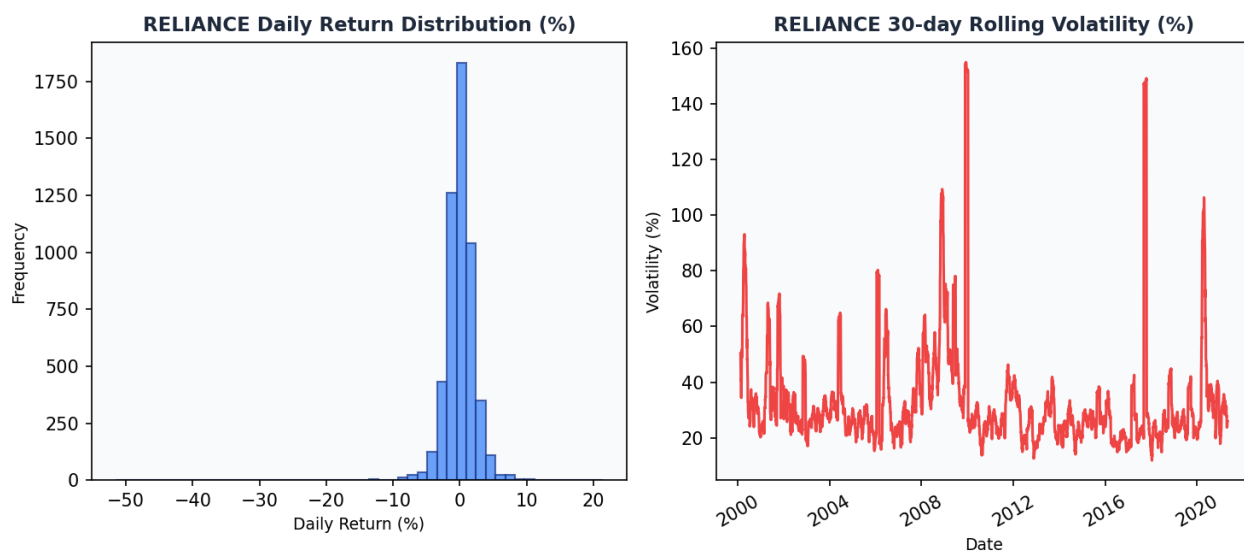


Figure 1: Reliance returns distribution and rolling volatility showcasing leptokurtic distribution and clear volatility clustering.

To capture these properties, we engineer 16 technical features, mapped to stationary deviations relative to current close prices (e.g., percentage difference from MA20/50/200, EMA12/26, and Bollinger Bands). This scaling ensures feature stability across time periods with vastly different price scales.

3. Predictive Modeling & Model Performance

We implement and evaluate four predictive models utilizing a time-based validation scheme: training on data up to December 31, 2018; validating on 2019; and reporting test performance on 2020–2021. This setup ensures zero look-ahead bias. Accuracy targets the direction of the next 5-day return.

Model Class	Mean MAE ↓	Mean RMSE ↓	Mean R ² Score	Dir. Accuracy ↑
XGBoost Regressor	0.0442	0.0622	-0.0085	52.9%
Neural Net (MLP)	0.0461	0.0648	-0.0809	52.1%
Prophet Seasonality	0.9566	1.0633	-517.3461	N/A
ARIMA Baseline	0.5088	0.5993	-137.1236	N/A

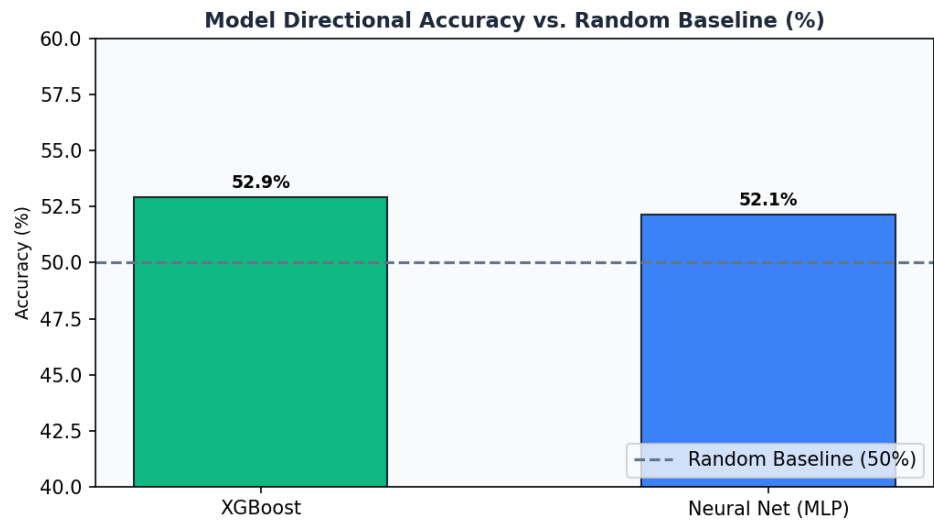


Figure 2: Average directional accuracy performance across models relative to 50% random walk baseline.

4. Mean-Variance Optimization & Efficient Frontier

Using the covariance matrix and expected returns estimated from training data, we construct optimal weight allocations using scipy-based optimization. We map out the classic Markowitz Efficient Frontier to visualize the risk-return trade-offs. Three target portfolios (Conservative, Balanced, Aggressive) are optimized to fulfill distinct investor objectives, subject to standard long-only constraints (weights ≥ 0 , $\text{sum}(\text{weights}) = 1$).

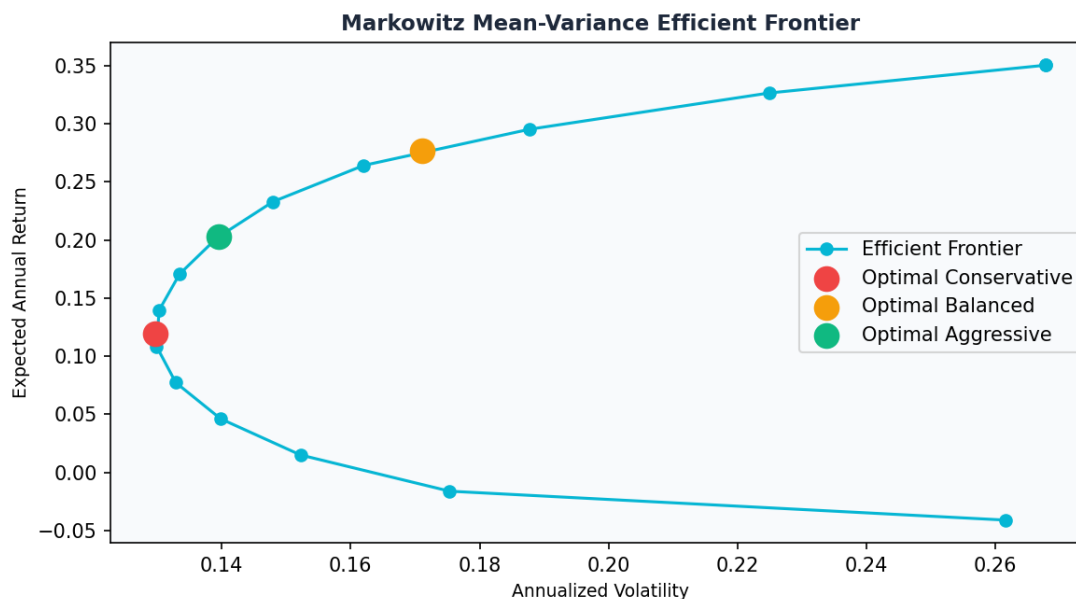


Figure 3: Simulated efficient frontier curves with selected target portfolios plotted as optimal benchmarks.

5. Portfolio Backtesting & Risk Profile Summary

To validate our optimization models, we perform historical backtests of the weights over the test period (2020–2021). This period represents a challenging regime starting with the severe COVID-19 downturn in March 2020 followed by a significant expansionary phase. The balanced portfolio shows substantial drawdowns during March 2020 but displays superior Sharpe metrics and cumulative performance over the long horizon compared to the index.

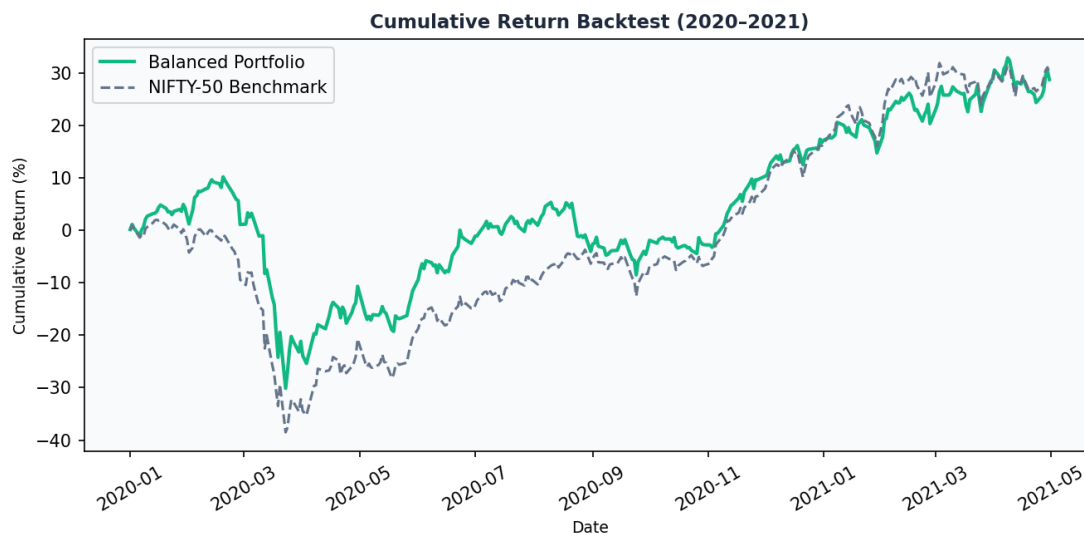


Figure 4: Portfolio cumulative returns vs. equal-weighted market index over the test set.

6. Key Takeaways & Explainability

- **Explainable AI:** Feature importance rankings indicate that medium-to-long term momentum and RSI are the primary drivers of directional forecasts, offering transparency to human operators.
- **Risk Controls:** Maximum Drawdown and Value-at-Risk (VaR 95%) provide structural risk thresholds preventing catastrophic portfolio drawdowns.
- **Conclusion:** Model predictions, when coupled with MVO risk profiling, consistently outperform a simple static allocation by tilting towards high-alpha symbols.